



The Four Operations Word Problems: Gold

1. Eleanor has 73 marbles. Benji has 28 more marbles than Eleanor. They combine their marbles then share them evenly between them. Work out the how many marbles they will each get.
2. Sajid receives £132 for his birthday. He spends £46.50 on clothes, £39.80 on a computer game and saves the rest. Work out how much money Sajid saves.
3. Ethan has 19 bags of beads. Each bag contains 17 beads, except one bag which contains 32 beads. Work out the total number of beads that Ethan has.
4. Mai has a bag containing 216 sweets. She wants to share these sweets equally between 12 friends. How many sweets will each friend receive?
5. In 2016, Alexia used 3967 units of electricity. In 2017, Alexia used 4050 units of electricity. Each unit costs 12p. Work out how much more money Alexia spent on electricity in 2017 compared to 2016.
6. A car park has 135 free spaces on the ground floor, 109 free spaces on the first floor and 72 free spaces on the second floor. 37 cars leave and 45 cars enter the car park. Work out the total number of free spaces in the car park now.
7. A pack of 6 rulers costs £1.18. Mr Jones needs to buy enough rulers for 30 students. How much money will Mr Jones spend on rulers?
8. Lewis has 14 075 photos stored in the cloud. His brother has 19 881 photos. Each photo takes up 2.5MB of storage. How much more storage space does Lewis' brother require to store his photos?
9. Paulo walks 3.75 miles per day. Work out the total number of miles that Paulo will walk in 4 weeks.
10. On a school trip, there are 29 teachers and 349 students. They travel on minibuses which can hold up to 15 people. Work out how many spare seats there will be.
11. There are 1900 students in a school. Each classroom can hold 35 students. Work out the minimum number of classrooms required to ensure each student is allocated to a room.
12. A pack of 14 biscuits costs £1.17. A cake costs £2.19. Arjun wants to buy 98 biscuits and 3 cakes. Work out how much money Arjun will spend.
13. Bella and three of her friends meet for lunch. Their total bill is £39.20 plus a 15% tip. Assuming they share the bill evenly, work out how much each person will pay.



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14. Amira is making bracelets. She spends £13.90 on materials, then sells 47 bracelets for £3.20 each. Work out how much profit Amira will make.
15. Romana buys packs of stickers. She buys 3 packs which each contain 40 stickers, 2 packs which each contain 27 stickers and 5 packs which each contain 12 stickers. She then sells each sticker for 35p. Work out the total amount of money she makes from selling all her stickers.

Challenge:

Jericho bought a car for £12 400. He pays a deposit of £850, then asks to pay the remaining amount off over 5 years. For this to happen, he must pay a 15% surcharge on the remaining amount. Work out how much money Jericho will pay each year.



The Four Operations Word Problems: Gold **Answers**

1. Eleanor has 73 marbles. Benji has 28 more marbles than Eleanor. They combine their marbles then share them evenly between them. Work out the how many marbles they will each get.

$$73 + 28 = 101$$

$$101 + 73 = 174$$

$$174 \div 2 = 87 \text{ marbles}$$

2. Sajid receives £132 for his birthday. He spends £46.50 on clothes, £39.80 on a computer game and saves the rest. Work out how much money Sajid saves.

$$132 - (46.50 + 39.80) = £45.70$$

3. Ethan has 19 bags of beads. Each bag contains 17 beads, except one bag which contains 32 beads. Work out the total number of beads that Ethan has.

$$18 \times 17 = 306$$

$$306 + 32 = 338 \text{ beads}$$

4. Mai has a bag containing 216 sweets. She wants to share these sweets equally between 12 friends. How many sweets will each friend receive?

$$216 \div 12 = 18 \text{ sweets}$$

5. In 2016, Alexia used 3967 units of electricity. In 2017, Alexia used 4050 units of electricity. Each unit costs 12p. Work out how much more money Alexia spent on electricity in 2017 compared to 2016.

$$4050 - 3967 = 83$$

$$83 \times 12 = 996\text{p or }£9.96$$

6. A car park has 135 free spaces on the ground floor, 109 free spaces on the first floor and 72 free spaces on the second floor. 37 cars leave and 45 cars enter the car park. Work out the total number of free spaces in the car park now.

$$135 + 109 + 72 + 37 - 45 = 308 \text{ spaces}$$

7. A pack of 6 rulers costs £1.18. Mr Jones needs to buy enough rulers for 30 students. How much money will Mr Jones spend on rulers?

$$30 \div 6 = 5$$

$$5 \times 1.18 = £5.90$$

8. Lewis has 14 075 photos stored in the cloud. His brother has 19 881 photos. Each photo takes up 2.5MB of storage. How much more storage space does Lewis' brother require to store his photos?

$$19\,881 - 14\,075 = 5806$$

$$5806 \times 2.5 = 14\,515\text{MB or }14.515\text{GB}$$

9. Paulo walks 3.75 miles per day. Work out the total number of miles that Paulo will walk in 4 weeks.

$$3.75 \times 7 \times 4 = 105 \text{ miles}$$

10. On a school trip, there are 29 teachers and 349 students. They travel on minibuses which can hold up to 15 people. Work out how many spare seats there will be.

$$29 + 349 = 378$$

$$378 \div 15 = 25 \text{ r } 3$$

They will need 26 minibuses. The last minibus will have 3 students on it, so there will be 12 spare seats.



11. There are 1900 students in a school. Each classroom can hold 35 students. Work out the minimum number of classrooms required to ensure each student is allocated to a room.

$$1900 \div 35 = 54 \text{ r } 10$$

The minimum number of classrooms is 55.

12. A pack of 14 biscuits costs £1.17. A cake costs £2.19. Arjun wants to buy 98 biscuits and 3 cakes. Work out how much money Arjun will spend.

$$98 \div 14 = 7$$

$$(7 \times 1.17) + (3 \times 2.19) = \text{£}14.76$$

13. Bella and three of her friends meet for lunch. Their total bill is £39.20 plus a 15% tip. Assuming they share the bill evenly, work out how much each person will pay.

$$15\% \text{ of } 39.20 = 5.88$$

$$39.20 + 5.88 = 45.08$$

$$45.08 \div 4 = \text{£}11.27$$

14. Amira is making bracelets. She spends £13.90 on materials, then sells 47 bracelets for £3.20 each. Work out how much profit Amira will make.

$$47 \times 3.20 = 150.40$$

$$150.40 - 13.90 = \text{£}136.50 \text{ profit}$$

15. Romana buys packs of stickers. She buys 3 packs which each contain 40 stickers, 2 packs which each contain 27 stickers and 5 packs which each contain 12 stickers. She then sells each sticker for 35p. Work out the total amount of money she makes from selling all her stickers.

$$3 \times 40 = 120$$

$$2 \times 27 = 54$$

$$5 \times 12 = 60$$

$$120 + 54 + 60 = 234 \text{ stickers}$$

$$234 \times 35 = 8190\text{p or } \text{£}81.90$$

Challenge:

Jericho bought a car for £12 400. He pays a deposit of £850, then asks to pay the remaining amount off over 5 years. For this to happen, he must pay a 15% surcharge on the remaining amount. Work out how much money Jericho will pay each year.

$$12\,400 - 850 = 11\,550$$

$$15\% \text{ of } 11\,550 = 1732.50$$

$$11\,550 + 1732.50 = 13\,282.50$$

$$13\,282.50 \div 5 = \text{£}2656.50 \text{ per year}$$